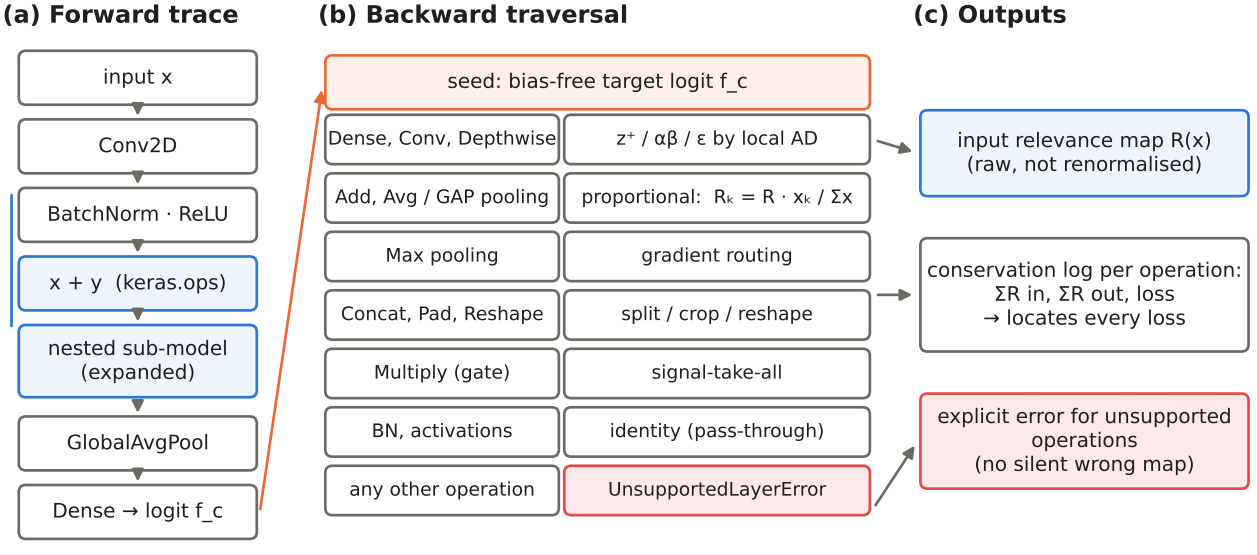
(a) Forward trace
input x
Conv2D
BatchNorm · ReLU
x + y (keras.ops)
nested sub-model (expanded)
GlobalAvgPool
Dense → logit f_c
(b) Backward traversal
seed: bias-free target logit f_c
Dense, Conv, Depthwise
z⁺ / αβ / ε by local AD
Add, Avg / GAP pooling
proportional: Rₖ = R · xₖ / Σx
Max pooling
gradient routing
Concat, Pad, Reshape
split / crop / reshape
Multiply (gate)
signal-take-all
BN, activations
identity (pass-through)
any other operation
UnsupportedLayerError
(c) Outputs
input relevance map R(x)
(raw, not renormalised)
conservation log per operation:
ΣR in, ΣR out, loss
→ locates every loss
explicit error for unsupported operations
(no silent wrong map)
blue: raw tensor ops and nested models are ordinary nodes of the traced graph
reverse topological order; R from several consumers is summed;
weighted ops: R = x ⊙ ∇(f(x; w̃), s), s = R_out / z